HALF ADDER & FULL ADDER

Digital Logic Design using Logisim

Half Adder | Full Adder | Truth Tables | Logisim Simulation

KLH-FED-2026-Section-2--Batch-8

github.com/Keerthan858/KLH-FED-2026-Section-2--Batch-

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1. What is a Binary Adder?

A combinational circuit that adds binary numbers.

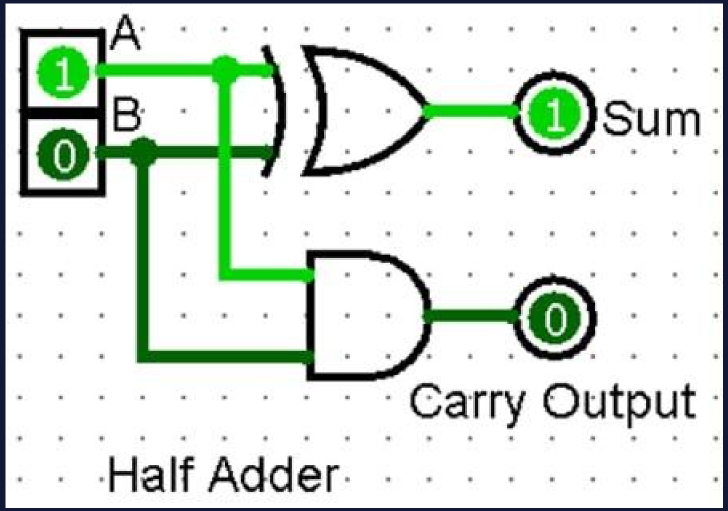
SUM is the result bit; CARRY is the overflow.

A core building block of ALUs, CPUs and calculators.

Two common circuits: Half Adder and Full Adder.

2. Half Adder

Adds two 1-bit binary inputs: A and B



Half Adder is a simple circuit that adds two single bit binary numbers.

Inputs: A, B

Outputs: Sum (S), Carry (C)

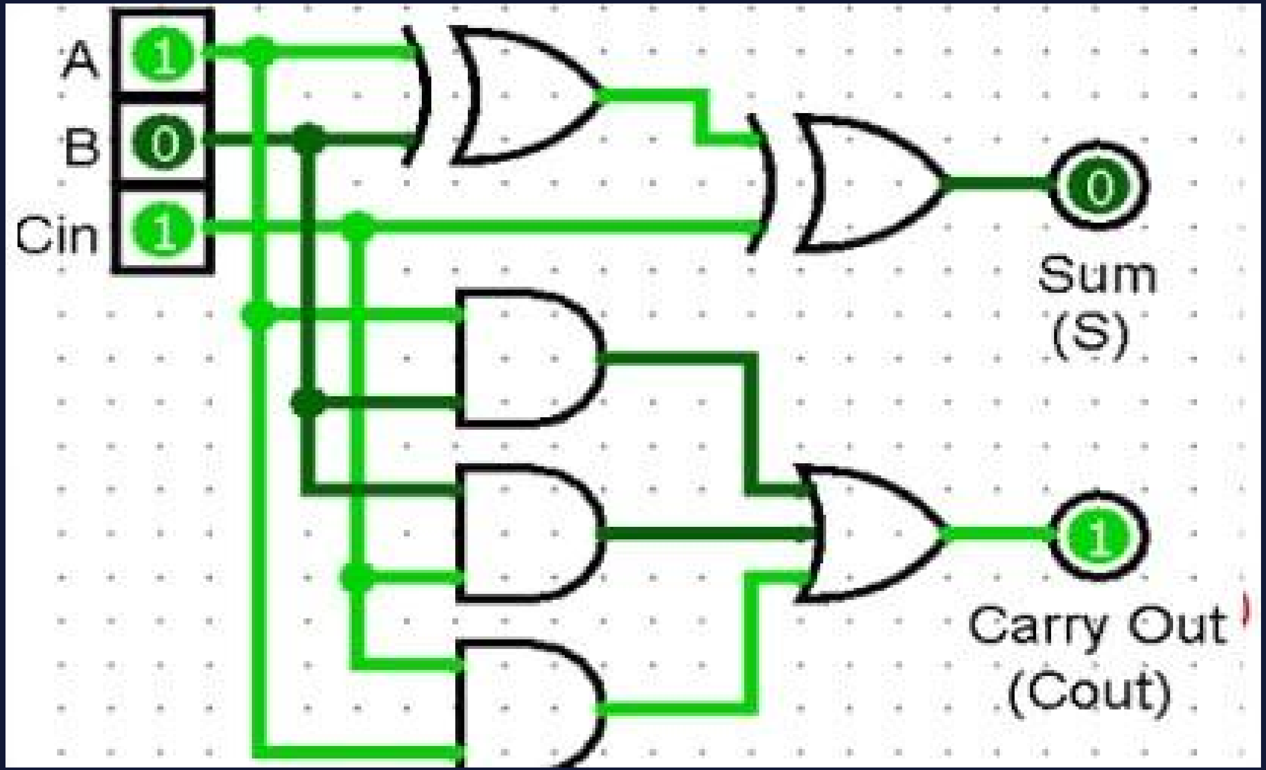
SUM is produced by XOR: $S = A \oplus B$

CARRY is produced by AND: $C = A \cdot B$

A	B	SUM ($A \oplus B$)	CARRY ($A \cdot B$)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

7. Full Adder using Two Half Adders

Two Half Adders + one OR gate — modular, easy to build, test and expand



A full Adder is an advanced circuit that adds three single bit binary numbers.

Inputs: A, B, Cin

Outputs: SUM and Carry-Out (Cout)

SUM = $A \oplus B \oplus Cin$

Cout = $AB + Cin(A \oplus B)$

Built from two Half Adders + one OR gate.

6. Full Adder Truth Table

A	B	Cin	SUM	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

THANK YOU

Questions?

Half Adder • Full Adder • Logisim